

EXP NO:13

Determination and plotting of intrinsic impedance and phase velocity of various medium

Enter μ_r value: 5×10^{-3}

Enter ϵ_r value: 6×10^{-4}

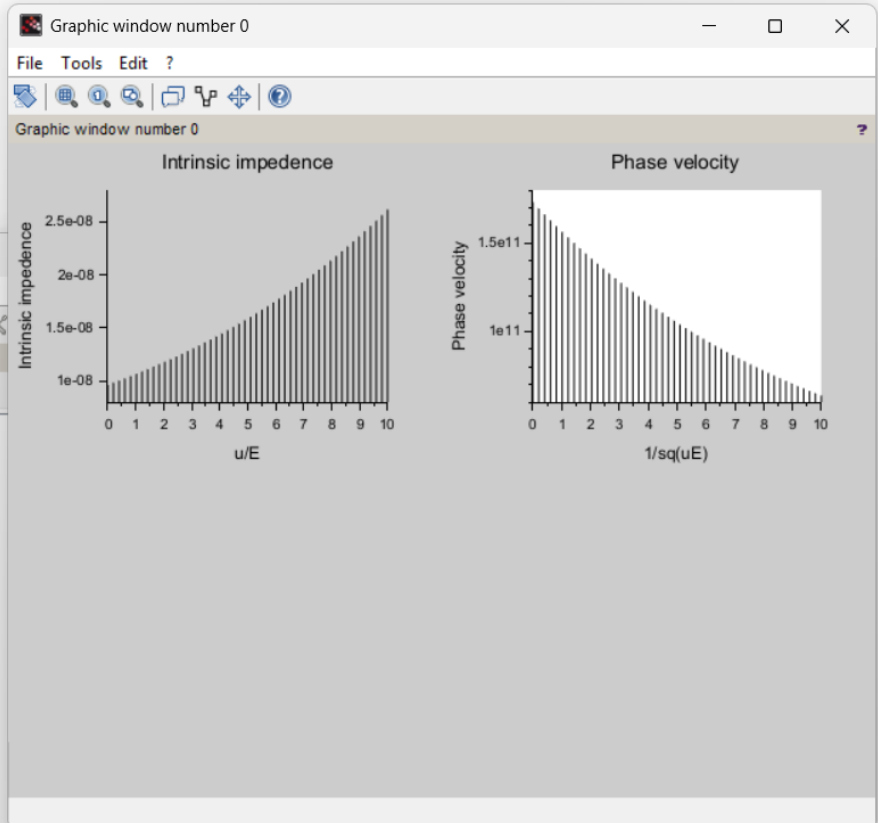
"Intrinsic impedance:"
9.627D-09
"Phase velocity:"
1.731D+11

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci)...

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exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

```
exp333.scd X
1 clc;
2 clear;
3 ur=input("Enter- $\mu_r$ -value:");
4 er=input("Enter- $\epsilon_r$ -value:");
5 uo=4*3.14*10^-7;
6 eo=8.854*10^-12;
7 n=sqrt(uo*ur/er*eo);
8 disp(["Intrinsic-impedance:"],[n]);
9 pv=1/sqrt(uo*ur*er*eo);
10 disp(["Phase-velocity:"],[pv]);
11 figure();
12 x=linspace(0,10,-50);
13 y=n*exp(0.1*x);
14 z=pv*exp(-0.1*x);
15 subplot(2,-2,-1);
16 plot2d3(x,y);
17 title("Intrinsic-impedance");
18 xlabel("u/E");
19 ylabel("Intrinsic-impedance");
20 subplot(2,-2,-2);
21 plot2d3(x,z);
22 title("Phase-velocity");
23 xlabel("1/sq(uE)");
24 ylabel("Phase-velocity");
25
```



Calculation:

$$Exp := \frac{13}{-}$$

$$\mu_r = 5 \times 10^{-3}$$

$$\epsilon_r = 6 \times 10^{-4}$$

$$\mu_0 = 4\pi \times 10^{-7}$$

$$= 4 \times 3.14 \times 10^{-7}$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

$$\mu_0 = 1.256 \times 10^{-6}$$

i) intrinsic impedance. $\Rightarrow \eta = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_r} \epsilon_0} \Rightarrow \sqrt{\frac{(1.256 \times 10^{-6})(5 \times 10^{-3})}{6 \times 10^{-4}} (8.854 \times 10^{-12})}$

$$= \sqrt{\frac{6.28 \times 10^{-9}}{6 \times 10^{-4}} \times (8.854 \times 10^{-12})} = \sqrt{(1.0467 \times 10^{-5}) (8.854 \times 10^{-12})}$$

$$\eta = \sqrt{9.269 \times 10^{-17}} \Rightarrow \eta = 9.627 \times 10^{-9}$$

ii) Phase velocity calculation

$$v_p = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_r \epsilon_0}}$$

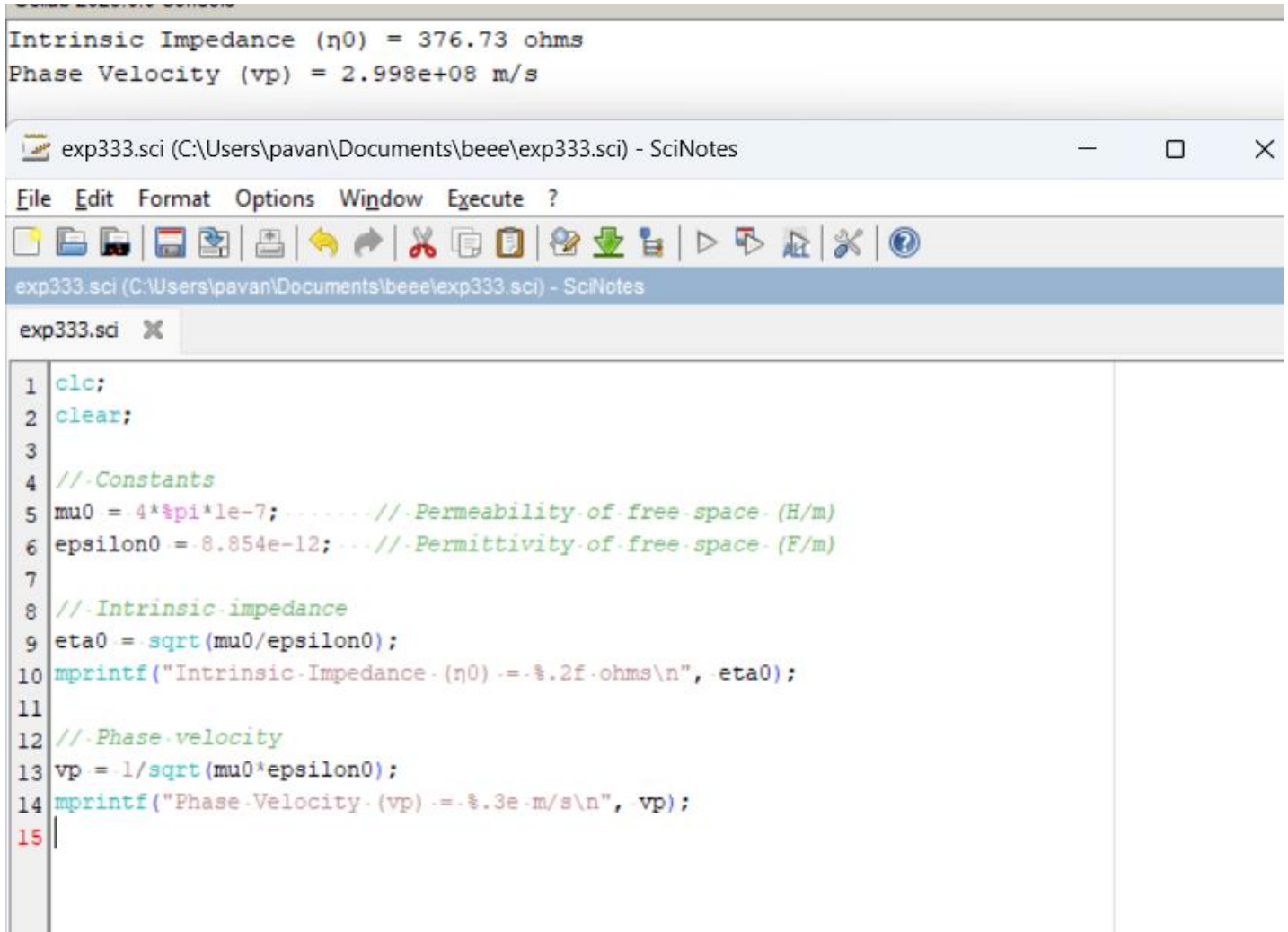
$$v_p = \sqrt{(1.256 \times 10^{-6})(5 \times 10^{-3})(6 \times 10^{-4})(8.854 \times 10^{-12})}$$

$$v_p = \frac{1}{\sqrt{(6.28 \times 10^{-9})(5.3124 \times 10^{-15})}} \Rightarrow v_p = \frac{1}{\sqrt{3.334 \times 10^{-23}}} \Rightarrow v_p = 1.731 \times 10^{11} \text{ m/s}$$

CASE:1

To determine the intrinsic impedance and phase velocity of electromagnetic waves in free space (vacuum).

```
Intrinsic Impedance ( $\eta_0$ ) = 376.73 ohms
Phase Velocity (vp) = 2.998e+08 m/s
```

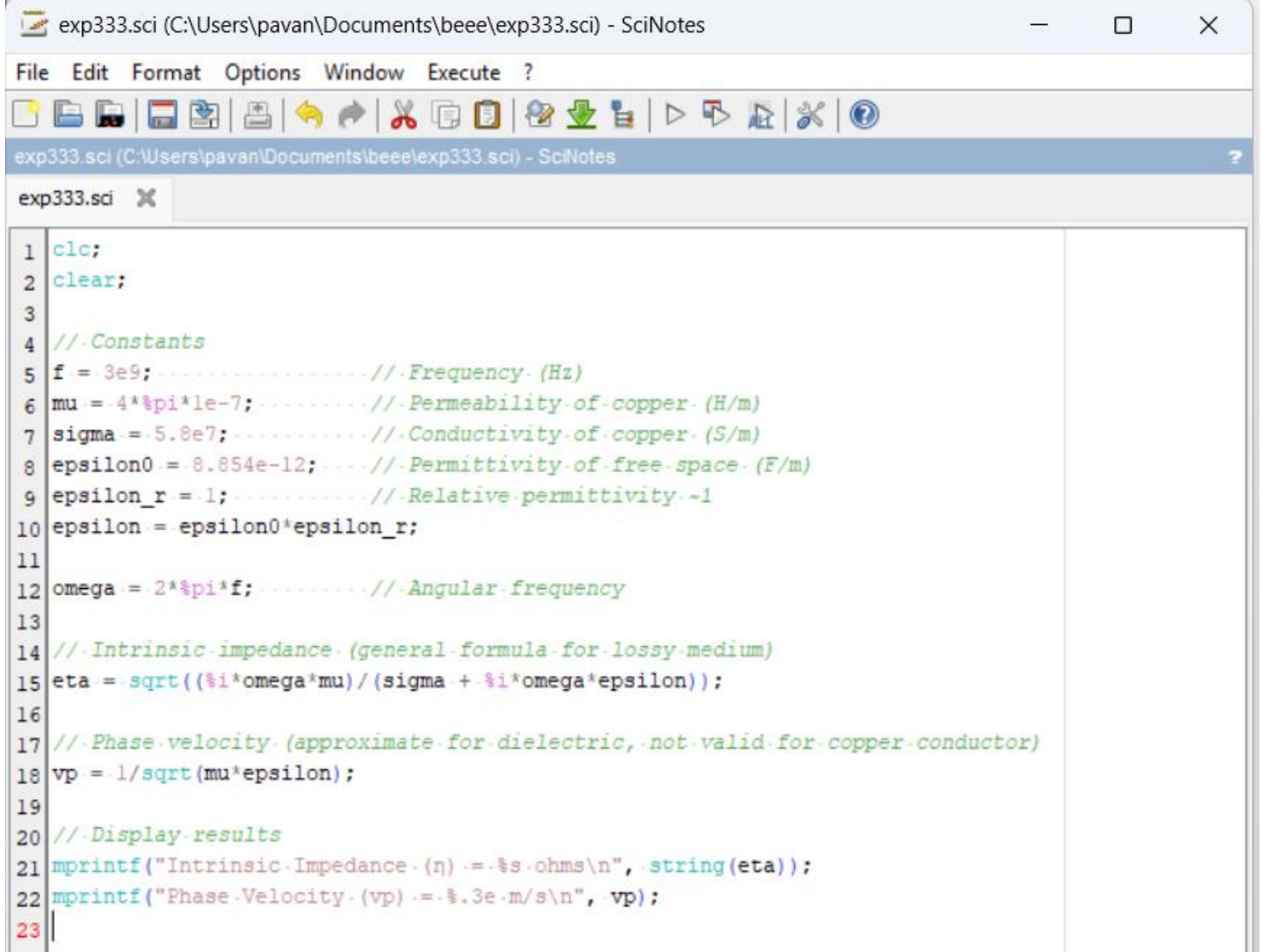


The screenshot shows a SciNotes window titled "exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes". The window contains MATLAB code that calculates the intrinsic impedance and phase velocity in free space. The code is as follows:

```
1 clc;
2 clear;
3
4 //Constants
5 mu0 = 4*pi*1e-7; .....//Permeability of free space (H/m)
6 epsilon0 = 8.854e-12; ...//Permittivity of free space (F/m)
7
8 //Intrinsic impedance
9 eta0 = sqrt(mu0/epsilon0);
10 mprintf("Intrinsic Impedance - ( $\eta_0$ ) = %.2f-ohms\n", eta0);
11
12 //Phase velocity
13 vp = 1/sqrt(mu0*epsilon0);
14 mprintf("Phase Velocity (vp) = %.3e-m/s\n", vp);
15
```

CASE:2 To determine the intrinsic impedance and phase velocity for a wave propagating through a conductive medium like copper.

Intrinsic Impedance (η) = 0.0142898+ i *0.0142898 ohms
Phase Velocity (vp) = 2.998e+08 m/s



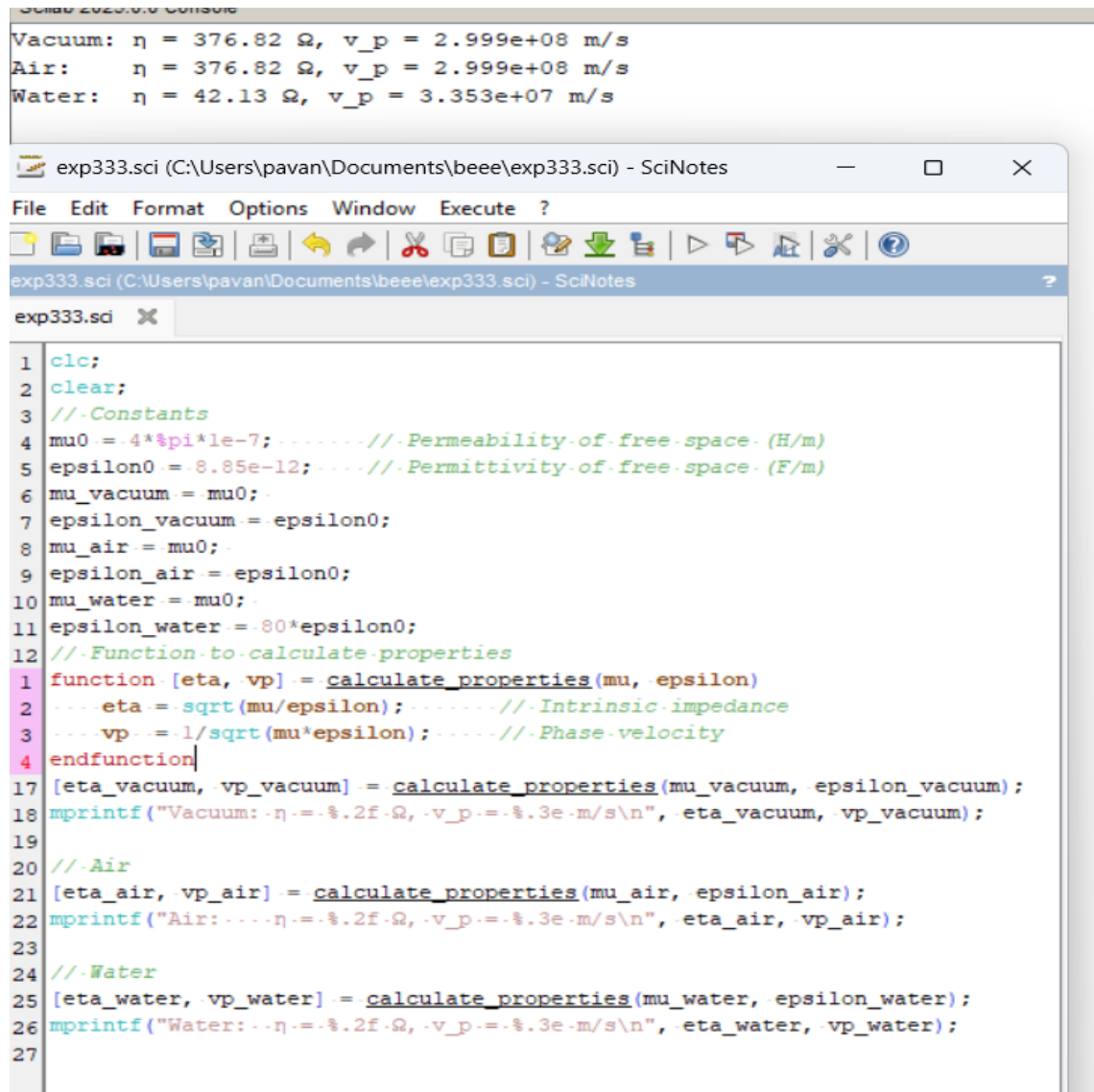
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exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci X
1 clc;
2 clear;
3
4 // Constants
5 f = 3e9; .....// Frequency (Hz)
6 mu = 4*pi*1e-7; .....// Permeability of copper (H/m)
7 sigma = 5.8e7; .....// Conductivity of copper (S/m)
8 epsilon0 = 8.854e-12; .....// Permittivity of free space (F/m)
9 epsilon_r = 1; .....// Relative permittivity ~1
10 epsilon = epsilon0*epsilon_r;
11
12 omega = 2*pi*f; .....// Angular frequency
13
14 // Intrinsic impedance (general formula for lossy medium)
15 eta = sqrt((i*omega*mu)/(sigma + i*omega*epsilon));
16
17 // Phase velocity (approximate for dielectric, not valid for copper conductor)
18 vp = 1/sqrt(mu*epsilon);
19
20 // Display results
21 mprintf("Intrinsic Impedance (eta) = %s ohms\n", string(eta));
22 mprintf("Phase Velocity (vp) = %3e m/s\n", vp);
23
```

case 3: To determine and compare the intrinsic impedance and phase velocity of electromagnetic waves propagating through different media: Vacuum, Air, and Water.

```

Vacuum:  $\eta = 376.82 \, \Omega$ ,  $v_p = 2.999e+08 \, \text{m/s}$ 
Air:     $\eta = 376.82 \, \Omega$ ,  $v_p = 2.999e+08 \, \text{m/s}$ 
Water:   $\eta = 42.13 \, \Omega$ ,  $v_p = 3.353e+07 \, \text{m/s}$ 

```



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File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci
1 clc;
2 clear;
3 %%-Constants
4 mu0 = 4*pi*1e-7; % Permeability of free space (H/m)
5 epsilon0 = 8.85e-12; % Permittivity of free space (F/m)
6 mu_vacuum = mu0;
7 epsilon_vacuum = epsilon0;
8 mu_air = mu0;
9 epsilon_air = epsilon0;
10 mu_water = mu0;
11 epsilon_water = 80*epsilon0;
12 %%-Function to calculate properties
13 function [eta, vp] = calculate_properties(mu, epsilon)
14     eta = sqrt(mu/epsilon); % Intrinsic impedance
15     vp = 1/sqrt(mu*epsilon); % Phase velocity
16 endfunction
17 [eta_vacuum, vp_vacuum] = calculate_properties(mu_vacuum, epsilon_vacuum);
18 fprintf("Vacuum: -eta = %.2f-ohm, -v_p = %.3e-m/s\n", eta_vacuum, vp_vacuum);
19
20 %%-Air
21 [eta_air, vp_air] = calculate_properties(mu_air, epsilon_air);
22 fprintf("Air: -eta = %.2f-ohm, -v_p = %.3e-m/s\n", eta_air, vp_air);
23
24 %%-Water
25 [eta_water, vp_water] = calculate_properties(mu_water, epsilon_water);
26 fprintf("Water: -eta = %.2f-ohm, -v_p = %.3e-m/s\n", eta_water, vp_water);
27

```

